The Rytova–Keldysh interaction is a positive Gaussian mixture: kernel-checked admissibility for planar universal optimality and Keldysh-jellium Madelung energies of TMD Wigner lattices

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Abstract

In units $e^2/(4\pi\varepsilon_0) = 1$, the Rytova–Keldysh potential of a freestanding two-dimensional semiconductor is $V(r) = \frac{\pi}{2r_0}[\mathbf{H}_0 - Y_0](r/r_0)$. We give a human proof ([C], not machine-checked) that, as a function of $t = r^2$, it is a positive mixture of Gaussians with weight $w(a) = r_0^{-1} e^{a/r_0^2} \operatorname{erfc}(\sqrt{a}/r_0)$. It is therefore completely monotone in t, which is the class covered by the openai/math theorem on universal optimality of the triangular lattice [6].

Tier A (kernel-checked in Lean 4.34.1). The axiom footprint is {`propext`, `Classical.choice`, `Quot.sound`} and planted controls are caught. We prove a generic lemma: every positive exponential mixture with finite moments satisfies upstream's `AdmissiblePotential`. For the mixture-*defined* Keldysh potential we prove admissibility, strict positivity, and a Gaussian-regularised variant. Applying upstream's `universal_energy_minimum` then gives the triangular-lattice inequalities. Every triangular-lattice statement is conditional on that upstream theorem, which is AI-generated and not peer reviewed, and on the fidelity of its definitions. The unregularised inequality is expected to read $\top \leq \top$ [C]. Finiteness of the regularised energy is not proved [C].

Tier C. That the mixture equals the Struve form is a human proof, given here step by step.

Tier X. Floating-point checks only. A new rusty-SUNDIALS module computes Keldysh-jellium Madelung energies of the triangular and square Wigner lattices by an Ewald split in the mixture variable. It reproduces the Coulomb values at $r_0 = 0$ and the Dedekind-η log-gas gap as $r_0 \to \infty$. It finds $E_{\mathrm{sq}} - E_{\mathrm{tri}} > 0$ at all 51 sampled $r_0\sqrt{n}$. For MoS_2, $MoSe_2$, WS_2 and WSe_2 in vacuum we take r_0 from published polarizabilities [L]. A HuggingFace C2DB-derived dataset is used to identify the 2H phases and to cross-check lattice constants. It contains no polarizabilities.

None of this is a breakthrough. Complete monotonicity of V follows quickly from textbook integrals. What is new is the machine-checked admissibility against upstream's definition, and the tiered bookkeeping.

Tiers. [A]: kernel-checked Lean, with axioms within the allowlist and planted controls caught. [L]: a cited source, quoted in the ledger (prose as extracted, mathematics transcribed linearly, elisions marked). This is looser than the strict Elenchus `citation_check`, which accepts only theorem-kind

statements quoted verbatim: KT-L-0001, KT-L-0002 and KT-L-0005 (an equation, table values and a README) and the Remark 1.5 numbers in KT-L-0003 would be refused on kind, and the linearised mathematics would not match verbatim. Sandier–Serfaty Theorem 2 passes that strict check (Stage-4 audit). [C]: an argument or human proof that is not machine-checked. [X]: floating-point exploration, which never supports a claim. Ledger identifiers (`KT-A-0006` and so on) refer to `ledger/ledger.json` (Section 7). A Tier A tag attaches to the Lean statement, never to its physical gloss.

1 Setting

Berkelbach, Hybertsen and Reichman [1] write the in-plane interaction of charges in a monolayer transition-metal dichalcogenide (TMD) between dielectrics $\varepsilon_1, \varepsilon_2$ as $V_{2D}(\rho) = \frac{\pi e^2}{(\varepsilon_1+\varepsilon_2)\rho_0}[\mathbf{H}_0(\rho/\rho_0) - Y_0(\rho/\rho_0)]$. They call it "a form derived by Keldysh" [2]. In vacuum the screening length is $\rho_0 = 2\pi\chi_{2D}$, after Cudazzo et al. [3] [L, KT-L-0001]. With $\varepsilon_1 = \varepsilon_2 = 1$ and $e^2/(4\pi\varepsilon_0) = 1$ this is

$$V(r) = \frac{\pi}{2r_0}\,\mathbf{K}_0(r/r_0), \qquad \mathbf{K}_0 = \mathbf{H}_0 - Y_0 \text{ (DLMF 11.2.5 [5])}, \qquad \widehat{V}(k) = \frac{2\pi}{k(1+r_0k)} \tag{1}$$

[C, KT-C-0004]. The Fourier transform uses $\widehat{V}(k) = \int_{\mathbb{R}^2} V(|x|)e^{-ik\cdot x}\,d^2x$, and the limit $r_0 \to 0$ recovers the Coulomb kernel $2\pi/k$.

Upstream [6] calls $g : \mathbb{R} \to \mathbb{R}$ *admissible* if it is C^∞ on $(0,\infty)$, $g \geq 0$ there, and $(-1)^n g^{(n)} \geq 0$ for all n. The pair interaction is $g(|x-y|^2)$, so g is a function of the *squared* distance. Upstream proves that the triangular lattice A of density one has the least lower energy among locally finite density-one configurations. Energies are taken in $[0,\infty]$ (Listing 1) [A, KT-A-0001].

Listing 1: Upstream definition and theorem (openai/math, pinned copy; proof elided).

```
def AdmissiblePotential (g : ℝ → ℝ) : Prop :=
  ContDiffOn ℝ (⊤ : ℕ∞) g (Set.Ioi 0) ∧
  (∀ t : ℝ, 0 < t → 0 ≤ g t) ∧
  ∀ (r : ℕ) (t : ℝ), 0 < t → 0 ≤ (-1 : ℝ) ^ r * iteratedDeriv r g t

theorem universal_energy_minimum (g : ℝ → ℝ) (C : Set Plane)
    (hg : AdmissiblePotential g) (_hC : LocallyFinite C) (hd : DensityOne C) :
    latticeEnergy g ≤ energy g C ∧ latticeEnergy g = energy g A := -- proof elided
```

`gen_inputs.py` extracts the listings from the pinned sources into `gen/*.lean`, byte for byte. The typeset `gen/*.lst` copies differ only in that each non-ASCII symbol is escaped to its LaTeX glyph.

2 Step-by-step human demonstration

This section is written for a strong undergraduate. Each step gives the mathematical statement and its proof, then the Lean declaration that checks it. Steps 0–6 are mirrored in Lean [A]. Step 7, the link to the Struve function, is not formalized; it is [C].

Step 0: the weight. For $x \geq 0$ put $\operatorname{erfcx}(x) = \frac{2}{\sqrt{\pi}}\int_0^\infty e^{-(y^2+2xy)}\,dy$. Substituting $s = y + x$ gives $\operatorname{erfcx}(x) = e^{x^2}\frac{2}{\sqrt{\pi}}\int_x^\infty e^{-s^2}ds = e^{x^2}\operatorname{erfc}(x)$, the usual scaled complementary error function. Lean *defines* `erfcx` by the integral, because Mathlib at this pin has no erfc. The two forms agree numerically: the max relative error is 1.34×10^{-31} at seven points, and a perturbed integrand ($2xy \to xy$) is off by at least 5.6×10^{-3} at the six points with $x > 0$; at $x = 0$ the perturbation has no effect [X, KT-X-0001]. For $r_0 > 0$ set $w(a) = r_0^{-1}\operatorname{erfcx}(\sqrt{a}/r_0)$.

Lemma 1 (bounds on the weight)**.** *For $x \geq 0$, $0 < \operatorname{erfcx}(x) \leq 1$. For $x > 0$, $\operatorname{erfcx}(x) \leq 1/(\sqrt{\pi}\, x)$. Hence, for $a > 0$, $0 < w(a) \leq 1/r_0$ and $w(a) \leq 1/(\sqrt{\pi}\sqrt{a})$.*

Proof. The integrand is positive and continuous, so $\operatorname{erfcx}(x) > 0$. For $y > 0$ and $x \geq 0$ we have $y^2 + 2xy \geq y^2$, so the integral is at most $\int_0^\infty e^{-y^2} dy = \sqrt{\pi}/2$, which gives $\operatorname{erfcx} \leq 1$. If $x > 0$, drop y^2 instead: $\int_0^\infty e^{-2xy} dy = 1/(2x)$, so $\operatorname{erfcx}(x) \leq \frac{2}{\sqrt{\pi}} \cdot \frac{1}{2x}$. For w, substitute $x = \sqrt{a}/r_0$: $w \leq r_0^{-1} \cdot r_0/(\sqrt{\pi}\sqrt{a})$. □

Lean: `erfcx_le_one`, `erfcx_le_inv`, `erfcx_pos`, `w_le`, `w_le_sqrt` [A, KT-A-0005].

Step 1: every moment is finite.

Lemma 2. *For $r_0 > 0$, $n \in \mathbb{N}$ and $c > 0$, the function $F_n(a) = (4a)^{-n} e^{-c/(4a)}\, w(a)/(2a)$ is integrable on $(0, \infty)$.*

Proof. Split at $a = 1$. *On $(0, 1]$:* put $u = 1/(4a)$, so that $w(a)/(2a) = 2u\, w(a)$ and $F_n = 2w(a)\, u^{n+1} e^{-cu}$. From $e^{cu} \geq (cu)^N/N!$ we get $u^N e^{-cu} \leq N!/c^N$. So $F_n \leq 2r_0^{-1}(n+1)!/c^{n+1}$, a constant, and a bounded measurable function on an interval of finite length is integrable. *On $(1, \infty)$:* here $u \leq 1/4 \leq 1$, so $u^n \leq 1$ and $e^{-cu} \leq 1$. Then $F_n \leq w(a)/(2a) \leq \frac{1}{2\sqrt{\pi}} a^{-3/2}$ by Lemma 1, and $\int_1^\infty a^{-3/2} da = 2 < \infty$. □

Lean: `pow_mul_exp_neg_le` (the bound $u^N e^{-cu} \leq N!/c^N$) and `keldysh_integrable`, which uses exactly the split $(0, 1] \cup (1, \infty)$ [A, KT-A-0005].

Step 2: the generic lemma. Let (α, μ) be a measure space. Let k, m be measurable with $k \geq 0$, $m \geq 0$ μ-a.e., and suppose $\int k^n e^{-ck} m \, d\mu < \infty$ for all n and all $c > 0$. Put $G_n(t) = \int (-k)^n e^{-tk} m \, d\mu$ and $g = G_0$.

Theorem 3. *g is admissible, and $g^{(n)}(t) = G_n(t)$ for $t > 0$.*

Proof. Fix $t > 0$ and let s range over $(t/2, \infty)$, a neighbourhood of t. The s-derivative of $(-k)^n e^{-sk} m$ is $(-k)^{n+1} e^{-sk} m$. Its absolute value is $k^{n+1} e^{-sk} m \leq k^{n+1} e^{-(t/2)k} m$, because $s \geq t/2$ and $k \geq 0$. The right side does not depend on s, and it is integrable by hypothesis with $c = t/2$. So the theorem on differentiation under the integral sign (dominated convergence) gives $G_n'(t) = G_{n+1}(t)$. Induction on n then shows $G_0 \in C^N(0, \infty)$ for every N, and $g^{(n)} = G_n$ on $(0, \infty)$. Finally $(-1)^n G_n(t) = \int k^n e^{-tk} m \, d\mu \geq 0$, since the integrand is nonnegative a.e.; $n = 0$ gives $g \geq 0$. □

Lean: `hasDerivAt_G` (via Mathlib's `hasDerivAt_integral_of_dominated_loc_of_deriv_le` on `Ioi (t/2)`), `contDiffOn_G`, `iteratedDeriv_G`, `G_sign`, `laplace_admissible`, `laplace_iteratedDeriv` [A, KT-A-0003]. Nonnegativity of m is essential. The Lean negative control `signedMix_not_admissible` shows that $e^{-t} - 2e^{-2t}$, a mixture with a negative weight, is not admissible, since it is negative at $t = \log \frac{3}{2}$. The positive control `expNeg_admissible` is the Dirac instance [A, KT-A-0004].

Step 3: the Keldysh instance. Take μ to be Lebesgue measure on $(0, \infty)$, $k(a) = 1/(4a)$ and $m(a) = w(a)/(2a)$. Lemma 2 is exactly the moment hypothesis, and $e^{-tk(a)} m(a) = \frac{w(a)}{2a} e^{-t/(4a)}$. So Theorem 3 applies to

$$\mathcal{K}_{r_0}(t) = \int_0^\infty \frac{w(a)}{2a}\, e^{-t/(4a)}\, da, \tag{2}$$

and $(-1)^n \mathcal{K}_{r_0}^{(n)}(t) = \int_0^\infty (4a)^{-n} \frac{w(a)}{2a} e^{-t/(4a)} da \geq 0$. Lean: `keldysh_data`, `keldysh_admissible` [A, KT-A-0006].

Step 4: non-vacuity. In Lean the Bochner integral of a non-integrable function is 0, and 0 is admissible. So admissibility alone could be vacuous. But for $t > 0$ the integrand in (2) is strictly positive on $(0, \infty)$ and integrable (Lemma 2 with $n = 0$, $c = t$). A nonnegative integrable function whose support has positive measure has positive integral, so $\mathcal{K}_{r_0}(t) > 0$. Lean: `keldysh_pos` [A, KT-A-0006].

Step 5: products. If f, h are admissible, then by Leibniz's rule

$$(-1)^n (fh)^{(n)} = \textstyle\sum_j \binom{n}{j} \left[(-1)^j f^{(j)}\right] \left[(-1)^{n-j} h^{(n-j)}\right] \geq 0,$$

so fh is admissible. With $h = e^{-\varepsilon t}$ ($\varepsilon \geq 0$, Step 2), the Gaussian-regularised $\mathcal{K}_{r_0}(t) e^{-\varepsilon t}$ is admissible. Lean: `admissible_mul` (H2 lane, imported) and `keldyshReg_admissible` [A, KT-A-0002, KT-A-0008].

Step 6: the triangular-lattice corollaries. Upstream's theorem gives, for every locally finite density-one C, latticeEnergy$(\mathcal{K}_{r_0}) \leq$ energy$(\mathcal{K}_{r_0}, C)$ and the same for the regularised interaction. These are `triangular_keldysh_optimal` and `triangular_keldyshReg_optimal` [A, KT-A-0007, KT-A-0008], conditional on upstream. They need careful reading. $V(r) \sim 1/r$ at large r, and $\sum_{0<|x|\leq R} 1/|x|$ grows like R over a planar lattice. So latticeEnergy$(\mathcal{K}_{r_0}) = \top$ is expected, and the first inequality then reads $\top \leq \top$ [C, KT-C-0002]. It is neither proved nor refuted in Lean. For $\varepsilon > 0$ the regularised sum should be finite [C, KT-C-0003], which would make the second inequality informative. Lean does not prove that finiteness. The physically relevant neutralised (jellium) energy is not covered by either statement (Section 5).

Step 7: the mixture is the Rytova–Keldysh potential [C, KT-C-0001].

Theorem 4 (not machine-checked)**.** *For $r_0 > 0$ and $r > 0$, $V(r) = \mathcal{K}_{r_0}(r^2)$, with V as in* (1)*.*

Proof. Every integrand below is nonnegative, so all interchanges of integrals are justified by Tonelli's theorem. This route was suggested by the Stage-4 audit (an AI agent, not a human referee).

(a) A Laplace form in r. DLMF 11.5.2 with $\nu = 0$ gives $\mathbf{K}_0(z) = \frac{2}{\pi} \int_0^\infty e^{-zu} (1 + u^2)^{-1/2} du$ for $\Re z > 0$ [L, KT-L-0004]. With $z = r/r_0$,

$$V(r) = \frac{1}{r_0} \int_0^\infty e^{-ru/r_0} \, (1 + u^2)^{-1/2} \, du.$$

This alone shows that V is completely monotone in r.

(b) Subordination. For $\rho > 0$ and $t > 0$, $e^{-\rho\sqrt{t}} = \int_0^\infty e^{-ta} \frac{\rho}{2\sqrt{\pi}} a^{-3/2} e^{-\rho^2/(4a)} \, da$. This is the Laplace transform of the Lévy density, a standard formula that we use without a citation. Take $\rho = u/r_0$ and $t = r^2$, insert it into (a), and exchange the u and a integrals (Tonelli):

$$V(r) = \int_0^\infty e^{-ta} \, \frac{a^{-3/2}}{2\sqrt{\pi}\, r_0^2} \left[\int_0^\infty \frac{u}{\sqrt{1+u^2}} e^{-\lambda u^2} du \right] da, \qquad \lambda = \frac{1}{4 r_0^2 a}.$$

(c) The inner integral is elementary. Substitute $v = \sqrt{1 + u^2}$, so that $dv = u(1 + u^2)^{-1/2} du$:

$$\int_0^\infty \frac{u}{\sqrt{1+u^2}} e^{-\lambda u^2} du = e^\lambda \int_1^\infty e^{-\lambda v^2} dv = \frac{1}{2} \sqrt{\frac{\pi}{\lambda}} \, e^\lambda \operatorname{erfc}(\sqrt{\lambda}) = \frac{1}{2} \sqrt{\frac{\pi}{\lambda}} \operatorname{erfcx}(\sqrt{\lambda}).$$

With $\sqrt{\pi/\lambda} = 2 r_0 \sqrt{\pi a}$ this gives $V(r) = \int_0^\infty e^{-ta} \rho(a) \, da$ with $\rho(a) = \dfrac{\operatorname{erfcx}\big(1/(2r_0\sqrt{a})\big)}{2 r_0 a}$.

(d) Change of variable. Put $a = 1/(4b)$, so $da = db/(4b^2)$ in absolute value and $e^{-ta} = e^{-t/(4b)}$. Then $\rho(a)\,da = \frac{4b\,\mathrm{erfcx}(\sqrt{b}/r_0)}{2r_0} \cdot \frac{db}{4b^2} = \frac{w(b)}{2b}\,db$, hence $V(r) = \int_0^\infty \frac{w(b)}{2b} e^{-r^2/(4b)} db = \mathcal{K}_{r_0}(r^2)$.

(e) A second route, kept as a cross-check. One can also invert the radial Fourier transform of $\widehat{V}$. Here $V \notin L^1(\mathbb{R}^2)$ and $\widehat{V} \notin L^1(\mathbb{R}^2)$, so the inversion $V(r) = \frac{1}{2\pi}\int_0^\infty \widehat{V}(k) J_0(kr) k\,dk = \int_0^\infty \frac{J_0(kr)}{1+r_0 k} dk$ is improper (distributional) and is not justified here. It then uses three standard integrals, none of which we cite: the table integral $\int_0^\infty J_0(bk)(k+c)^{-1} dk = \frac{\pi}{2}\mathbf{K}_0(bc)$ (checked numerically in the prior round, quadosc at $r \in \{0.1, 1, 5\}$); the Laplace pair $\int_0^\infty e^{-sa} e^{b^2 a}\,\mathrm{erfc}(b\sqrt{a})\,da = 1/(\sqrt{s}(\sqrt{s}+b))$, which gives $\frac{1}{1+r_0 k} = k\int_0^\infty w(a) e^{-ak^2} da$ (numerically to 1.9×10^{-41}, prior round [X]); and Weber's integral $\int_0^\infty e^{-ak^2} J_0(kr) k\,dk = e^{-r^2/(4a)}/(2a)$. J_0 changes sign, so the swap needs Fubini. Integrating $|J_0|$ in a first gives $\int_0^\infty |J_0(kr)|/(1+r_0 k)\,dk < \infty$, since $|J_0(x)| \le C\min(1, x^{-1/2})$. Route (a)–(d) needs none of this. □

Numerically, the Rust mixture, the Rust DLMF quadrature and mpmath's closed form agree on 28 points (Section 5) [X, KT-X-0002]. In mpmath at 30 digits, the Struve form, the Laplace form (a), the weight form (c) and the mixture (d) agree at four (r_0, r) points, and the inner integral (c) and the subordination formula (b) hold at the sampled parameters, all to a maximum relative error of 2.1×10^{-17}. Two planted errors are caught: the inner integrand with $e^{-\lambda u}$ in place of $e^{-\lambda u^2}$ misses the closed form of (c) by at least 0.37, and subordination with $a^{-1/2}$ in place of $a^{-3/2}$ misses $e^{-\rho\sqrt{t}}$ by at least 0.82 (relative) [X, KT-X-0010]. Theorem 4 is the definition-fidelity gap. The Lean theorem is about (2), and the reader must accept Theorem 4 to transfer it to $\mathbf{K}_0$.

3 Lean formalization

File: `formal_cert/Autoresearch/KeldyshCM.lean`, module `Autoresearch.KeldyshCM`. It imports `Autoresearch.Yukawa` (H2 lane) for `admissible_mul` and `admissible_congr`. Toolchain: pinned openai/math, Lean 4.34.1, Mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612`. Listings 2–5 are extracted verbatim by `gen_inputs.py`; proofs are elided.

Listing 2: Definitions (verbatim).

```
structure MixtureData (μ : Measure α) (k m : α → ℝ) : Prop where
  k_meas : Measurable k
  m_meas : Measurable m
  k_nonneg : ∀ᵐ x ∂μ, 0 ≤ k x
  m_nonneg : ∀ᵐ x ∂μ, 0 ≤ m x
  integrable : ∀ (n : ℕ) (c : ℝ), 0 < c → Integrable (fun x => k x ^ n * Real.exp (-(c * k x)) *
     m x) μ

noncomputable def erfcx (x : ℝ) : ℝ :=
  2 / Real.sqrt Real.pi * ∫ y in Ioi (0 : ℝ), Real.exp (-(y ^ 2 + 2 * x * y))

noncomputable def w (r0 a : ℝ) : ℝ := 1 / r0 * erfcx (Real.sqrt a / r0)

noncomputable def keldysh (r0 t : ℝ) : ℝ :=
  ∫ a in Ioi (0 : ℝ), w r0 a / (2 * a) * Real.exp (-t / (4 * a))

noncomputable def keldyshReg (r0 ε t : ℝ) : ℝ := keldysh r0 t * Real.exp (-(ε * t))

noncomputable def signedMix (t : ℝ) : ℝ := Real.exp (-t) - 2 * Real.exp (-(2 * t))
```

Listing 3: Generic lemma and its controls (statements verbatim).

```
theorem laplace_admissible (h : MixtureData μ k m) :
    AdmissiblePotential (fun t => ∫ x, Real.exp (-(t * k x)) * m x ∂μ) := -- proof elided

theorem laplace_iteratedDeriv (h : MixtureData μ k m) (n : ℕ) {t : ℝ} (ht : 0 < t) :
    iteratedDeriv n (fun t => ∫ x, Real.exp (-(t * k x)) * m x ∂μ) t =
      ∫ x, (-(k x)) ^ n * Real.exp (-(t * k x)) * m x ∂μ := -- proof elided

theorem expNeg_admissible {ε : ℝ} (hε : 0 ≤ ε) : AdmissiblePotential (fun t => Real.exp (-(ε * t)
    )) := -- proof elided

theorem signedMix_not_admissible : ¬ AdmissiblePotential signedMix := -- proof elided
```

Listing 4: The Keldysh instance (statements verbatim).

```
lemma erfcx_le_one {x : ℝ} (hx : 0 ≤ x) : erfcx x ≤ 1 := -- proof elided

lemma erfcx_le_inv {x : ℝ} (hx : 0 < x) : erfcx x ≤ 1 / (Real.sqrt Real.pi * x) := -- proof
    elided

lemma w_le {r0 : ℝ} (hr0 : 0 < r0) (a : ℝ) : w r0 a ≤ 1 / r0 := -- proof elided

lemma w_le_sqrt {r0 a : ℝ} (hr0 : 0 < r0) (ha : 0 < a) :
    w r0 a ≤ 1 / (Real.sqrt Real.pi * Real.sqrt a) := -- proof elided

lemma keldysh_integrable {r0 : ℝ} (hr0 : 0 < r0) (n : ℕ) {c : ℝ} (hc : 0 < c) :
    IntegrableOn (fun a => (1 / (4 * a)) ^ n * Real.exp (-(c * (1 / (4 * a)))) * (w r0 a / (2 * a
    )))
      (Ioi 0) := -- proof elided

lemma hasDerivAt_G (h : MixtureData μ k m) (n : ℕ) {t : ℝ} (ht : 0 < t) :
    HasDerivAt (G μ k m n) (G μ k m (n + 1) t) t := -- proof elided

lemma G_sign (h : MixtureData μ k m) (n : ℕ) (t : ℝ) : 0 ≤ (-1 : ℝ) ^ n * G μ k m n t := --
    proof elided

theorem keldysh_admissible {r0 : ℝ} (hr0 : 0 < r0) : AdmissiblePotential (keldysh r0) := -- proof
     elided

theorem keldysh_pos {r0 t : ℝ} (hr0 : 0 < r0) (ht : 0 < t) : 0 < keldysh r0 t := -- proof elided
```

Listing 5: Corollaries via upstream (statements verbatim).

```
theorem triangular_keldysh_optimal {r0 : ℝ} (hr0 : 0 < r0) (C : Set Plane) (hC : LocallyFinite C)
    (hdens : DensityOne C) : latticeEnergy (keldysh r0) ≤ energy (keldysh r0) C := -- proof
    elided

theorem keldyshReg_admissible {r0 ε : ℝ} (hr0 : 0 < r0) (hε : 0 ≤ ε) :
    AdmissiblePotential (keldyshReg r0 ε) := -- proof elided

theorem triangular_keldyshReg_optimal {r0 ε : ℝ} (hr0 : 0 < r0) (hε : 0 ≤ ε) (C : Set Plane)
    (hC : LocallyFinite C) (hdens : DensityOne C) :
    latticeEnergy (keldyshReg r0 ε) ≤ energy (keldyshReg r0 ε) C := -- proof elided
```

Axiom audit. The producer's compile exits with rc 0 in 23.9 s and has no warnings. Its `#print axioms` output is reproduced verbatim:

```
rc = 0, 23.9 s
'AutoresearchKeldysh.laplace_admissible' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchKeldysh.laplace_iteratedDeriv' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchKeldysh.expNeg_admissible' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchKeldysh.signedMix_not_admissible' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchKeldysh.erfcx_le_one' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchKeldysh.erfcx_le_inv' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchKeldysh.keldysh_integrable' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchKeldysh.keldysh_admissible' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchKeldysh.keldysh_pos' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchKeldysh.triangular_keldysh_optimal' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchKeldysh.keldyshReg_admissible' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchKeldysh.triangular_keldyshReg_optimal' depends on axioms: [propext, Classical.choice, Quot.
    sound]
```

At the paper stage, a `grep` on the claimed file for the words `sorry` and `admit`, and for `axiom` at the start of a line, finds no match: there are no `sorry`, no `admit` and no `axiom` declarations. The only occurrences of the word are the `#print axioms` commands. The paper-stage recompile, run by a separate agent instance, has status: **rc 0 in 848.1 s, axiom output identical to the producer's** (see Section 9).

Planted controls. `KeldyshCMControls.lean` is a separate file that imports the built module. It plants the false claim `AdmissiblePotential signedMix` twice:

```
theorem ctl_sorry : AdmissiblePotential signedMix := sorry
axiom ctl_cheat : False
theorem ctl_axiom : AdmissiblePotential signedMix := ctl_cheat.elim
```

Both compile with rc 0. Exit status is therefore not the gate; the axiom list is. Both are caught [A, KT-A-0009]:

```
rc = 0, 17.3 s
formal_cert/Autoresearch/KeldyshCMControls.lean:11:8: warning: declaration uses `sorry`
'AutoresearchKeldyshControls.ctl_sorry' depends on axioms: [propext, sorryAx, Classical.choice, Quot.sound]
'AutoresearchKeldyshControls.ctl_axiom' depends on axioms: [propext,
 AutoresearchKeldyshControls.ctl_cheat,
 Classical.choice,
 Quot.sound]
'AutoresearchKeldysh.laplace_admissible' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchKeldysh.keldysh_admissible' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchKeldysh.triangular_keldysh_optimal' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchKeldysh.keldyshReg_admissible' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchKeldysh.signedMix_not_admissible' depends on axioms: [propext, Classical.choice, Quot.sound]
```

What is and is not formalized. *Formalized:* Steps 0–6, with the weight defined through `erfcx`. *Not formalized:* Theorem 4, because Mathlib at this pin has no Struve or Neumann functions; latticeEnergy $= \top$ for $\mathcal{K}_{r_0}$; finiteness for the regularised interaction; and anything about jellium energies.

4 Physics: what the theorem does and does not say

System. Point charges of one sign in a freestanding TMD monolayer ($\varepsilon_1 = \varepsilon_2 = 1$), interacting through (1). Lengths are in units where the density is $n = 1$. A physical density n enters through $r_0\sqrt{n}$, since $E(n, r_0) = \sqrt{n}\,E(1, r_0\sqrt{n})$ in units $e^2/(4\pi\varepsilon_0) = 1$. The code checks this scaling to 9.7×10^{-16} [X, KT-X-0004]. *Regime.* Static classical lattice (Madelung) energies: no kinetic, exchange or correlation energy, no zero-point motion, no substrate.

What Tier A says. Take the mixture-defined interaction (2), or its Gaussian regularisation, as a function of squared distance. Among all locally finite configurations of density one, the triangular lattice attains the least lower energy in $[0, \infty]$, conditional on upstream [6].

What it does not say.

- For the raw Keldysh interaction the statement is expected to be $\top \leq \top$ [C].
- The ground state of electrons in a real TMD depends on quantum and correlation effects and on the dielectric environment. Nothing here bears on them.
- The neutralised energy that governs a Wigner crystal is not covered. Upstream's manuscript states a renormalised and jellium corollary for Riesz kernels with $0 \leq s < 2$, which is not in its Lean code (prior ledger UO-L-0007, UO-C-0003).
- No statement is made about where a Wigner crystal forms.

5 rusty-SUNDIALS numerics: Keldysh-jellium Madelung energies

Everything in this section is [X] and lies outside the Lean theorem.

Method. The code is the module `crates/qf-yukawa-crystal/src/keldysh.rs` and the binary `keldysh_jellium`, in the rusty-SUNDIALS worktree at commit `2e41505cc9df8c46d2b2d667c89ea53e977218ca` (branch `physics/ar-keldysh-2026-10-25`, *not pushed*). $V(r)$ is computed two ways:

- by DLMF 11.5.2 with adaptive Gauss–Kronrod 7–15;
- by the Gaussian mixture (2), the Lean definition.

The Ewald split is made in the mixture variable at $a^* = 1/(4\alpha^2)$. Gaussians with $a < a^*$ are summed in real space and the rest in reciprocal space. At $r_0 = 0$ this reduces exactly to the erfc split of the existing Coulomb module `jellium.rs`. Per particle at density ρ,

$$e = \tfrac{1}{2} \sum_{v \neq 0} V_{\mathrm{sr}}(|v|) + \tfrac{\rho}{2} \sum_{G \neq 0} \widehat{V}_{\mathrm{lr}}(G) - \tfrac{1}{2} V_{\mathrm{lr}}(0) - \tfrac{\rho}{2} \widehat{V}_{\mathrm{sr}}(0),$$

with $\widehat{V}_{\mathrm{sr}}(0) = 2\pi[r_0(\mathrm{erfcx}(X) - 1) + 2\sqrt{a^*/\pi}]$ and $X = \sqrt{a^*}/r_0$. The final term is the neutralising background. Because $\lim_{k \to 0}[2\pi/k - \widehat{V}(k)] = 2\pi r_0$, the jellium energy carries a lattice-independent shift $E_K - E_C = +\pi\rho r_0 + O(r_0^2)$ [C, KT-C-0006]. The structure-sensitive quantity is therefore the gap $E_{\mathrm{sq}} - E_{\mathrm{tri}}$.

Large-r_0 control. As $r_0 \to \infty$, $\widehat{V}(k) \sim r_0^{-1}\, 2\pi/k^2$, which is the 2D logarithmic kernel. Sandier and Serfaty give the renormalised log-gas energy of a lattice as $W(\Lambda) = -\frac{1}{2}\log(\sqrt{2\pi b}|\eta(\tau)|^2)$ (Lemma 3.3 of [4]). Their cells have area 2π. Rescaling to area 1 shifts W by a constant that is the same for every lattice, so the difference is unchanged and $r_0(E_{\mathrm{sq}} - E_{\mathrm{tri}}) \to \frac{1}{2}[q(e^{i\pi/3}) - q(i)]$ with $q(\tau) = \log(\sqrt{\Im\tau}|\eta(\tau)|^2)$ [C, KT-C-0005]. The code computes this value as 0.0052922513. It is consistent with their numerical values -0.2011 and -0.1958 (Remark 1.5) [L, KT-L-0003]. The computed r_0-scaled gaps converge to it at rate $1/r_0$, with no fitted constant (Table 2) [X, KT-X-0005].

Table 1: Checks ([X]; source `data/analysis.json`, macros generated by `gen_inputs.py`). + marks a positive control, − a negative control; unmarked rows are cross-checks.

Check	Result
Rust DLMF V vs mpmath $\frac{\pi}{2r_0}\mathbf{K}_0$, 28 points	max rel. err 1.8×10^{-16}
Rust mixture V (Lean definition) vs mpmath	2.4×10^{-15}
$\widehat{V}_{\mathrm{sr}} + \widehat{V}_{\mathrm{lr}} = 2\pi/(k(1 + r_0 k))$	1.8×10^{-15}
+ $r_0 = 0$: lattice Ewald vs `jellium.rs` $N = 288$ box (tri)	6.9×10^{-14}
+ $r_0 = 0$ vs ten-digit literature Madelung (tri / sq) [7, 8]	5.0×10^{-10} / 8.3×10^{-10}
+ $r_0 = 10^{-9}$: $E_K - E_C$ (required $< 10^{-8}$)	3.14×10^{-9}
+ $(E_K - E_C)/(\pi r_0)$ at $r_0 = 10^{-2}$ / 10^{-8}	0.98585 / 0.99999998
+ Ewald-parameter spread, $\alpha \in \{0.8, 1.6, 3.2\}$, $r_0 \in [0, 10^4]$	3.5×10^{-15}
− wrong-sign reciprocal half: spread (must be large)	$\geq 5.3 \times 10^{-2}$ (caught)
− neutralising background dropped: spread	≥ 0.216 (caught)
+ large r_0: $r_0(E_{\mathrm{sq}} - E_{\mathrm{tri}})$ vs $\frac{1}{2}[q(e^{i\pi/3}) - q(i)] = 0.0052922513$	rel. dev. $-3.4 \times 10^{-3} \to -3.4 \times 10^{-7}$

Table 2: $r_0(E_{\mathrm{sq}} - E_{\mathrm{tri}})$ at density one vs the Dedekind-η value 0.0052922513 ([X]).

r_0	$r_0(E_{\mathrm{sq}} - E_{\mathrm{tri}})$	relative deviation
10^2	0.0052744580	-3.4×10^{-3}
10^3	0.0052904675	-3.4×10^{-4}
10^4	0.0052920728	-3.4×10^{-5}
10^5	0.0052922334	-3.4×10^{-6}
10^6	0.0052922495	-3.4×10^{-7}

Result. At $r_0 = 0$, $E_{\mathrm{tri}} = -1.9605157893$ and $E_{\mathrm{sq}} = -1.9501324600$, a gap of 1.0383×10^{-2}. The gap is positive at all 51 sampled $r_0\sqrt{n} \in [10^{-3}, 10^2]$; the smallest value is 5.27×10^{-5}, at $r_0 = 100$ [X, KT-X-0006]. This compares two lattices only. It is not a proof and not a comparison with general configurations. Figure 1 shows the scan.

Materials. We take $r_0 = 2\pi\chi_{2D}$ with the DFT+RPA polarizabilities of Table I of [1]: 6.60, 8.23, 6.03 and 7.18 Å for MoS_2, $MoSe_2$, WS_2 and WSe_2. We also use the MoS_2 GW value of 5.0 Å [L, KT-L-0002]. Energies are converted with $e^2/(4\pi\varepsilon_0) = 14.3996454687$ eV Å, the scipy.constants value. Table 3 and Figure 2 give the results [X, KT-X-0007]. At $n = 10^{11}$ cm^{-2}, MoS_2 gives $E_{\mathrm{tri}} = -73.77$ meV and a gap of 0.431 meV. Relative to Coulomb at the same density, the magnitude of E_{tri} is lower mostly because of the lattice-independent $+\pi\rho r_0$ background term, which comes from the neutralisation convention rather than from lattice structure (as in Figs. 1 and 2). The structure-sensitive gap $E_{\mathrm{sq}} - E_{\mathrm{tri}}$ is also lower.

6 HuggingFace dataset illustration

Search. 23 Hub searches, 22 distinct terms (11 listed in `hf-data/keldysh_tmd/search.json` and 12 in `search2.txt`, with "c2db" repeated), found no dataset carrying a 2D polarizability α_{2D} or χ_{2D}. *Dataset used.* `colabfit/JARVIS_C2DB` at revision `432aeba9e505b59d0b3605c6346ac3816c754c02`, licence CC-BY-4.0. Its README cites Haastrup et al. [9] and ColabFit [10] [L, KT-L-0005]. Files read:

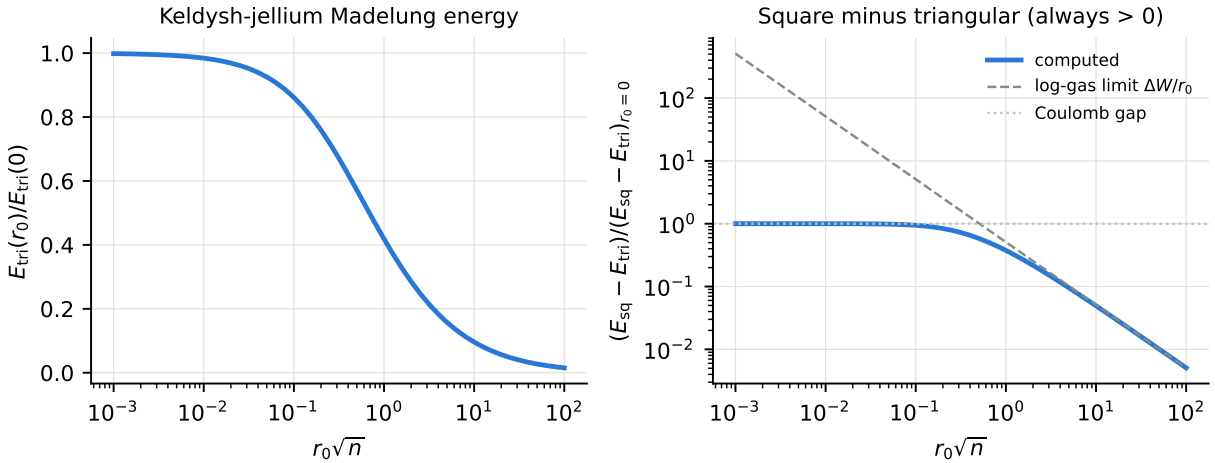


Figure 1: Keldysh-jellium Madelung energies at density one ([X]), generated by `scripts/keldysh/analyze.py` from `scan.json`. Left: $E_{\mathrm{tri}}(r_0)/E_{\mathrm{tri}}(0)$. This ratio is dominated by the lattice-independent $+\pi\rho r_0$ background term, which comes from the neutralisation convention, not from lattice structure. Right: the square–triangular gap, normalised by its Coulomb value, together with the log-gas asymptote $\Delta W/r_0$. The panel title "always > 0" is baked into the figure. It means positive at the 51 sampled points only.

Table 3: Vacuum Keldysh-jellium Madelung energies per electron at $n = 10^{12}$ cm^{-2} ([X]; r_0 [L]).

material	r_0 (Å)	$r_0\sqrt{n}$	E_{tri} (meV)	$E_{\mathrm{tri}}^{\mathrm{Coulomb}}$ (meV)	$E_{\mathrm{sq}} - E_{\mathrm{tri}}$ (meV)
MoS_2	41.47	0.415	-172.97	-282.31	0.961
$MoSe_2$	51.71	0.517	-159.17	-282.31	0.857
WS_2	37.89	0.379	-178.52	-282.31	1.002
WSe_2	45.11	0.451	-167.74	-282.31	0.922
MoS_2 (GW χ_{2D})	31.42	0.314	-189.72	-282.31	1.084

`README.md`, `ds.parquet`, `co/co_0.parquet` (1.98 MB). It has 3520 configurations, with DFT-PBE energy and band gap only.

What it shows. We use it only to *identify* the 2H monolayers: three sites, a hexagonal cell, and both chalcogens at the same in-plane point. We also cross-check lattice constants against the experimental values in Table II of [1] (Table 4) [X, KT-X-0008]. The PBE constants exceed experiment by 0.6–0.9%. The rows that pass the 2H test are the gapped ones, which is consistent with the 2H phase, but the gap was not the selection criterion.

Caveats. The dataset supplies no input to the energies; r_0 comes from [1]. C2DB itself publishes α_{2D}, but it is not on the Hub and was not downloaded.

7 Claim ledger

`ledger/build_ledger.py` files 30 claims: 9 at Tier A, 5 at L, 6 at C and 10 at X. Each claim has a content-addressed evidence blob in `ledger/evidence/`:

- Tier A: verbatim axiom lines and source hashes;
- Tier L: verbatim excerpts, in `ledger/sources/excerpts.json`;

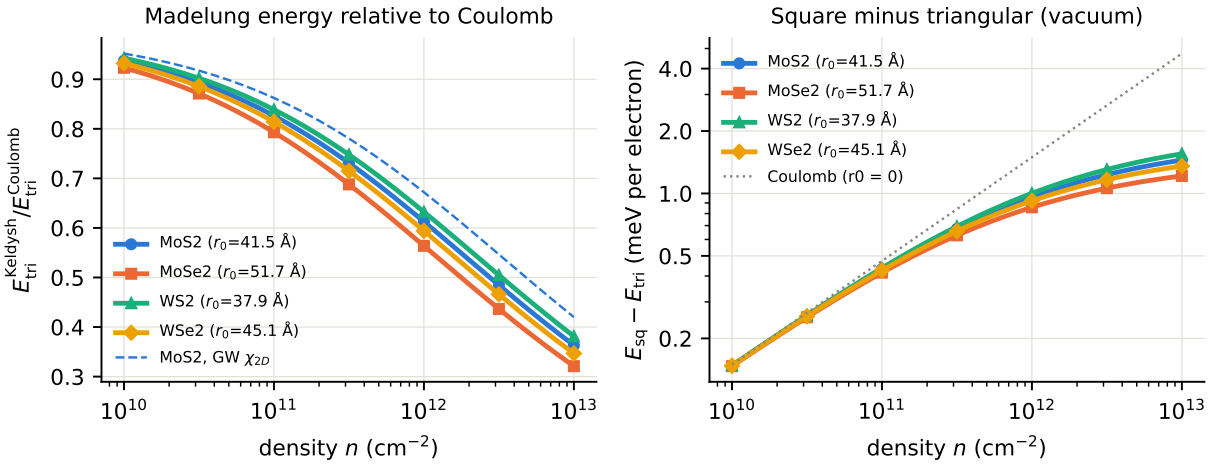


Figure 2: Material curves in vacuum ([X], with r_0 from [1] [L]). Left: E_{tri} relative to Coulomb. Like Fig. 1 (left), it is dominated by the $+\pi\rho r_0$ background term. Right: the square–triangular gap in meV per electron. The density range 10^{10}–10^{13} cm^{-2} is illustrative only. No substrate or hBN is modelled.

Table 4: 2H rows identified in `colabfit/JARVIS_C2DB` ([X]).

material	row	a PBE (Å)	a expt. [1] (Å)	PBE gap (eV)
MoS_2	`c2db_atoms_3238`	3.184	3.16	1.580
$MoSe_2$	`c2db_atoms_3150`	3.320	3.30	1.321
WS_2	`c2db_atoms_3223`	3.186	3.16	1.534
WSe_2	`c2db_atoms_3118`	3.319	3.29	1.238

- Tier C: argument text;
- Tier X: result files with `claim_licence: false`.

The Elenchus gate (`tools/ledger.py ledger.json --evidence-dir evidence`) exits with code 1. It reports **0 blocking findings** and **9 flags**, all `LEDGER_UNAUDITED_TIER_A`: no person has certified that the Tier A statements mean what their glosses say. The flags remain open.

8 Limitations

- **Upstream dependence.** Every triangular-lattice statement depends on openai/math's `universal_energy_minimum` (AI-generated, not peer reviewed) and on the fidelity of its definitions.
- **Definition fidelity.** `keldysh` is *defined* by the mixture. The identification with $\mathbf{H}_0 - Y_0$ (Theorem 4) is [C]. Its main route rests on DLMF 11.5.2 [L] and on the subordination formula, which we use without a citation and check only numerically.
- **Possibly empty corollaries.** The raw corollary is expected to be $\top \leq \top$. The regularised one lacks a finiteness proof.

- **Two lattices only.** The jellium numerics compare triangular with square, in vacuum, as static classical lattices.
- **Producer ≠ verifier is still open.** The paper-stage reruns (Section 9) were done by another AI agent instance in the same workflow. A Stage-4 audit by a further AI agent instance recompiled `Yukawa.lean` and `KeldyshCM.lean` into a private olean directory (rc 0, axioms within the allowlist, its own planted `sorry` and planted axiom caught), reran the scan (bit-identical) and the crate tests (28 passed, 0 failed, 1 ignored), and checked the citations (`AUDIT.md`). None of this is independent human verification, and the human proofs are unrefereed.
- **Novelty.** Complete monotonicity of V follows from textbook integrals (Step 7(a)). Our alphaXiv search did not find it stated, but that is weak evidence.
- **Dependency drift.** The imported `Yukawa.lean` belongs to another lane (sha256 below). If it changes, KeldyshCM must be rebuilt.

9 Reproducibility

AutoevolveAI files are uncommitted on top of base `db794dd98d12c2a79702eab5651ae7c36a35ea27` (branch `worktree-openai-math-discovery`). Source hashes (sha256):

- `KeldyshCM.lean`: `1bac0d410d366614c8d287f7ea95d495aa0ee09c4782cfa0d5b691a3048647f2`
- `KeldyshCMControls.lean`: `d7e5516fe00a6599a181d5a394dde70a66c533d0a2693b78b9035281801f0606`
- `Yukawa.lean`: `a16397de179837dd2119cf09c1b3086db7650b9b8bbaeb322f2d531936c033cf`

Rust: rusty-SUNDIALS commit `2e41505cc9df8c46d2b2d667c89ea53e977218ca`, not pushed.

```
taskset -c 4-7 python3 scripts/certified_numerics/lean_pinned.py compile formal_cert/Autoresearch/KeldyshCM.
    lean \
  --olean Autoresearch.KeldyshCM --json-out results/autoresearch/2026-10-25/H3/contribution/compile_keldysh.
    json
taskset -c 4-7 python3 scripts/certified_numerics/lean_pinned.py compile formal_cert/Autoresearch/
    KeldyshCMControls.lean \
  --json-out results/autoresearch/2026-10-25/H3/contribution/compile_controls.json
CARGO_TARGET_DIR=... cargo test --release -j 2 --offline --manifest-path <rusty-wt>/Cargo.toml -p qf-yukawa-
    crystal \
  -- --test-threads=2
keldysh_jellium scan results/keldysh/scan.json
python -I scripts/keldysh/analyze.py final results/keldysh <hf-data>/keldysh_tmd/tmd_rows.json
python -I papers/ar_keldysh_tmd/gen_inputs.py ; python -I papers/ar_keldysh_tmd/ledger/build_ledger.py
python3 <elenchus>/tools/ledger.py papers/ar_keldysh_tmd/ledger/ledger.json \
  --evidence-dir papers/ar_keldysh_tmd/ledger/evidence
```

Paper-stage reruns ([X, KT-X-0009]). These were run by a separate agent instance, not by a person:

- crate tests: 28 passed, 0 failed, 1 ignored (the ignored test predates this work);
- `keldysh_jellium scan`: output bit-identical to Stage 2's `scan.json`;
- Lean recompile of `KeldyshCM.lean`: rc 0 in 848.1 s, axiom output identical to the producer's.

References

[1] T. C. Berkelbach, M. S. Hybertsen, D. R. Reichman, *Theory of neutral and charged excitons in monolayer transition metal dichalcogenides*, arXiv:1305.4972v1 (2013).

[2] L. V. Keldysh, JETP Lett. **29**, 658 (1979). Cited as reference [26] of [1]; not read directly.

[3] P. Cudazzo, I. V. Tokatly, A. Rubio, Phys. Rev. B **84**, 085406 (2011). Cited as reference [27] of [1]; not read directly.

[4] E. Sandier, S. Serfaty, *From the Ginzburg–Landau model to vortex lattice problems*, arXiv:1011.4617v2 (2012).

[5] NIST Digital Library of Mathematical Functions, §11.2 (Eq. 11.2.5) and §11.5 (Eq. 11.5.2), `https://dlmf.nist.gov/11`, read 2026-10-25.

[6] OpenAI, *Universal optimality of the triangular lattice*, OpenAI Math release, github.com/openai/math, commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb` (2026). AI-generated; not peer reviewed. The commit id is inherited from the earlier published ledger; the pinned local copy has no `.git`, so it was not re-verified here.

[7] L. Bonsall, A. A. Maradudin, Phys. Rev. B **15**, 1959 (1977), doi:10.1103/PhysRevB.15.1959, as filed in the ledger of [11] (UO-L-0007).

[8] Alves et al., Phys. Rev. B **103**, 245125 (2021), doi:10.1103/PhysRevB.103.245125, as filed in the ledger of [11] (UO-L-0007).

[9] S. Haastrup et al. (C2DB), doi:10.1088/2053-1583/aacfc1. Cited from the dataset README; title and journal not verified by us.

[10] Haastrup et al., *JARVIS C2DB*, ColabFit (2023), doi:10.60732/37c26dae; HuggingFace `colabfit/JARVIS_C2DB`, revision `432aeba9e505b59d0b3605c6346ac3816c754c02`.

[11] X. Callens, *Planar universal optimality, tiered*, preprint v2, doi:10.5281/zenodo.23293429 (2026).

[12] X. Callens, *The triangular lattice and universal optimality* (textbook), doi:10.5281/zenodo.23295142 (2026).

This preprint applies the approach of [11, 12], kernel-checked consequences of upstream for explicit interactions, to the Rytova–Keldysh interaction.